

1. Administrative & Study Identification

Full Study Title: A Multi-center, Randomized, Double-blind, Parallel-group, 20-week Dose-finding Study to Evaluate Efficacy, Safety, and Tolerability of XXB750 in Patients With Resistant Hypertension **Short Title/Acronym:** An Efficacy, Safety, Tolerability and Dose Finding Study of XXB750 in Resistant Hypertension Patients **Protocol Number:** CXXB750B12201 **NCT Number:** NCT05562934 **Sponsor Name:** Novartis Pharmaceuticals / Novartis Pharma AG **Investigational Product:** XXB750 (Natriuretic Peptide Receptor-1 [NPR-1] Agonist) **Phase of Development:** Phase 2 **Medical Monitor:** Novartis Medical Monitor (Clinical Operations Lead)

2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To determine the dose-response relationship of XXB750 versus placebo in reducing the mean 24-hour ambulatory systolic blood pressure (SBP) from baseline to week 12 in patients with resistant hypertension. * **Secondary Objectives:** To evaluate the overall safety, tolerability, and effect of different doses of XXB750 on office and home blood pressure, as well as to assess target engagement via cyclic guanosine monophosphate (cGMP) levels.

2.2 Background & Rationale Resistant hypertension (rHTN) remains a significant clinical challenge where patients have uncontrolled blood pressure despite adherence to 3 or 4 guideline-directed antihypertensive agents. XXB750 is a novel NPR-1 agonist administered subcutaneously that activates natriuretic peptides, which regulate vascular tone, natriuresis, and diuresis. This study evaluates whether adding XXB750 to a maximally tolerated triple-drug background regimen effectively lowers blood pressure.

3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional * **Control Type:** Placebo-controlled * **Blinding:** Double-blind * **Randomization:** Randomized, Parallel-group

3.2 Planned Enrollment & Duration * **Target \$N\$:** Approximately 170 participants * **Number of Sites:** Multi-center, global * **Estimated Study Start:** 08-Nov-2022 * **Estimated Primary Completion:** 02-Jul-2024

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male and female participants ≥ 18 years old. 2. Apparent rHTN at screening, defined as uncontrolled BP with an office mean sitting systolic BP (msSBP) ≥ 140 mmHg despite treatment with stable, optimal, or maximally tolerated doses of three or four antihypertensive drugs of different classes (including an ACEI/ARB, a long-acting dihydropyridine CCB, and a thiazide/thiazide-like diuretic). 3. Mean 24-hour SBP ≥ 135 mmHg (measured by ABPM) at the end-of run-in visit.

4.2 Exclusion Criteria 1. Serum potassium > 5.0 mmol/L at screening or end-of-run-in visit. 2. Current therapy with a mineralocorticoid receptor antagonist (MRA) or sacubitril/valsartan, or usage within 4 weeks prior to screening. 3. Type I diabetes mellitus or uncontrolled Type II diabetes (HbA1c $\geq 9\%$). 4. Clinically significant cardiac arrhythmias, or any history of stroke, acute myocardial infarction, or PCI/CABG within 6-12 months of screening.

5. Intervention & Treatment Plan

5.1 Investigational Regimen * **Dosage/Frequency:** XXB750 (30 mg, 60 mg, 120 mg, or 240 mg) or matching placebo administered once every 4 weeks (Q4W) for a total of 3 doses (at baseline, Week 4, and Week 8). * **Route of Administration:** Subcutaneous (SC) injection. * **Duration of Treatment:** 12-week primary endpoint period within a 20-week overall study duration (plus a 2-week run-in period).

5.2 Concomitant & Prohibited Therapy * **Permitted:** Maximally tolerated background antihypertensive therapy (thiazide/thiazide-like diuretic, ACEi/ARB, and a long-acting dihydropyridine CCB). * **Prohibited:** Mineralocorticoid receptor antagonists (MRA), sacubitril/valsartan, more than 4 baseline antihypertensives, and prolonged use of NSAIDs (except low-dose aspirin).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening / Run-in	Weeks -2 to 0	Single-blind placebo injection, Baseline ABPM, medication adherence check
Baseline	Day 0	Randomization, First dose of XXB750 or placebo, Baseline office BP
Interim	Weeks 4 & 8	Office BP, Subsequent study drug SC injections
Primary Endpoint	Week 12	24-hour ABPM evaluation, Efficacy & Safety assessments
End of Study	Week 20	Safety follow-up, Exit Interview

(Note: Additional ambulatory monitoring occurs at Week 9, and home blood pressure is measured by participants on specified days throughout the study).

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * **Definition:** Dose-response relationship of XXB750 vs placebo in reducing the mean 24-hour ambulatory systolic blood pressure (SBP) from baseline to week 12. *

Measurement Method: 24-hour Ambulatory Blood Pressure Monitoring (ABPM).

7.2 Secondary & Exploratory Endpoints * Changes from baseline in office msSBP and home blood pressure. * Target engagement indicated by increases in cyclic guanosine monophosphate (cGMP) levels. * Changes in plasma renin levels and urinary sodium excretion.

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Standard collection via office visits and remote follow-ups up to Week 20 (8 weeks post-final dose).
 - **Serious Adverse Events (SAEs):** Routine monitoring and reporting within 24 hours per GCP.
 - **Data Monitoring Committee (DMC):** Internal safety monitoring mechanisms managed by Novartis.
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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Intent-to-Treat (ITT) for primary efficacy; Safety Set for AE evaluations.
 - **Sample Size Justification:** \$N \approx 170\$ powered to assess the dose-response curve across four escalating doses (30mg to 240mg) and placebo.
 - **Handling of Missing Data:** Mixed-effects Model for Repeated Measures (MMRM) or multiple imputation utilized for handling missing ABPM data over the 12-week timeframe.
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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
 - **Monitoring Plan:** Routine clinical onsite and remote monitoring.
 - **Audit Trail:** Blood pressure data captured via validated automated digital BP devices (e.g., Microlife Watch BP Office) and drug adherence tracked via drug assays from plasma samples.
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11. Study Identifiers & Governance

- **Primary ID:** CXXB750B12201
 - **Registry Number:** NCT05562934
 - **Sponsor/Lead Organization:** Novartis Pharmaceuticals
 - **Study Title:** An Efficacy, Safety, Tolerability and Dose Finding Study of XXB750 in Resistant Hypertension Patients
 - **Official Regulatory Title:** A Multi-center, Randomized, Double-blind, Parallel-group, 20-week Dose-finding Study to Evaluate Efficacy, Safety, and Tolerability of XXB750 in Patients With Resistant Hypertension
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12. Clinical Framework (Resistant Hypertension Specific)

- **Primary Condition:** Resistant Hypertension (rHTN)
 - **Diagnosis Threshold:** Office msSBP ≥ 140 mmHg and 24h ABPM SBP ≥ 135 mmHg
 - **Medical Methodology:** NPR-1 Agonism on top of triple background antihypertensive therapy
 - **Optimization Target:** 24-hour Ambulatory SBP reduction
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13. Study Procedures & Methodology

- **Management Pathway:** Triple/Quadruple guideline-directed antihypertensive therapy + Monthly SC XXB750
 - **Education Protocol (HCP):** Site training for protocol administration and ABPM device calibration
 - **Education Protocol (Patient):** Home blood pressure measurement instructions using provided automated devices
 - **Quality Audit Mechanism:** Assessment of background antihypertensive medication adherence via routine plasma drug assays
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14. Observation & Follow-Up Schedule

- **Total Duration:** 20 weeks (plus 2-week run-in)
 - **Onsite Visit Cadence:** Baseline, Weeks 4, 8, 12, and 20
 - **Remote Monitoring Frequency:** Intermittent home BP logging on specified days
 - **Checklist Review Interval:** ABPM completed at Baseline, Week 9, and Week 12
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15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				
Clinical Operations Lead	Laurie Heinrich (or Delegate)	____					

[DD-MMM-YYYY] | | *Lead Statistician* | [Name] | _____ | [DD-MMM-YYYY] |